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WHU-MARS: A Multispectral Aerial-Ground Benchmark Towards Any-Scenario Person Re-Identification

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https://github.com/msm8976/WHU-MARS

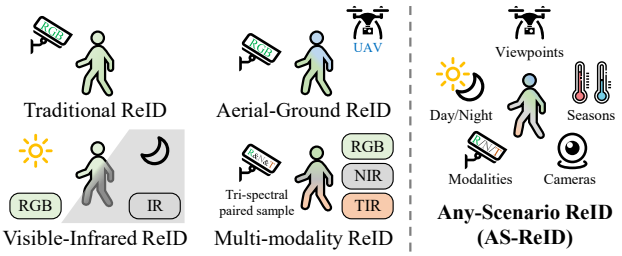
Abstract

*Recent person re-identification (ReID) leverages heterogeneous sensing with multiple modalities and viewpoints to improve robustness across diverse conditions. However, most approaches target predefined scenario pairs (e.g., visible-infrared or aerial-ground) and train separate task-specific models. In contrast, real-world systems often query against galleries that cover all scenarios, making such designs inefficient and complex to deploy. To bridge this gap, we introduce **Any-Scenario ReID (AS-ReID)**: given a query from any (modality, viewpoint) scenario, a single model retrieves the same identity from a heterogeneous gallery spanning all scenarios. Progress toward AS-ReID is limited by two factors: (i) the lack of a real-world-aligned benchmark with broad scenario coverage, and (ii) the challenge of learning representations with strong intra-identity cohesion and inter-identity discrimination under diverse scenarios. To this end, we construct **WHU-MARS**, a multispectral aerial-ground benchmark with 2,337 identities and 434,620 images captured by RGB, near-infrared, and thermal infrared cameras on both ground and UAV platforms. WHU-MARS spans day-night, multiple seasons, and diverse weather, enabling AS-ReID as well as conventional ReID tasks. We further propose the **Unified Alignment and Discrimination (UAD)** framework. Progressive Center Alignment (ProCA) aggregates multi-view features into modality centers and then aligns them toward identity centers to reduce scenario bias. Global Prototype Discrimination (GPD) contrasts samples against global identity prototypes to enforce large-margin discrimination. Extensive experiments highlight the challenges of WHU-MARS and demonstrate the effectiveness of UAD on AS-ReID.*

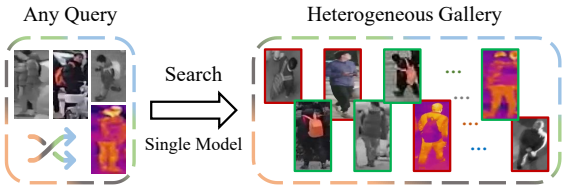
1. Introduction

Person re-identification (ReID) aims to retrieve a target individual across non-overlapping cameras and plays an important role in smart cities and public safety [38]. RGB-only systems can be unreliable under challenging illumination and environmental conditions [58], which motivates heterogeneous sensing across modalities and viewpoints: near-infrared (NIR) is effective in low light [40, 43], thermal infrared (TIR) remains informative under smoke or camouflage [59], and unmanned aerial vehicle (UAV) platforms provide wide-area coverage and flexible viewpoints beyond fixed cameras [53]. In modern deployments, ReID therefore must retrieve identities across heterogeneous sensing configurations and scenarios.

As illustrated in Figure 1, existing research branches into several task settings: Traditional ReID (Tr-ReID) focuses on ground RGB cameras; Aerial-Ground ReID (AG-



(a) Different ReID tasks and their corresponding scenarios.



(b) AS-ReID: Any-query to heterogeneous gallery retrieval.

Figure 1. Motivation and definition of Any-Scenario ReID (AS-ReID). (a) Existing ReID settings typically focus on *predefined scenario pairs*, while AS-ReID targets retrieval *across any (modality, viewpoint) scenario*. (b) Given any query (RGB/NIR/TIR, aerial/ground), a *single model* retrieves the same identity from a heterogeneous gallery spanning all scenarios.

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ReID) [28] targets cross-view matching; Visible-Infrared ReID (VI-ReID) [39] enables all-day matching across visible and infrared; and Multi-Modality ReID (MM-ReID) [59] leverages complementary spectra for robustness. While effective within their own settings, these tasks are organized around predefined scenario pairs (e.g., visible-infrared or aerial-ground) with task-specific models and evaluation protocols. In real-world applications, however, a query may come from any modality and any viewpoint, and the gallery typically contains all available scenarios. Such pairwise designs fragment retrieval across scenarios and hinder learning or evaluating a single model that works universally. This raises a fundamental question: ***I)*** *Can we move beyond predefined scenario pairs and establish a unified retrieval paradigm that works across any scenario?*

As summarized in Table 1, most existing datasets are built around predefined scenario pairs. VI-ReID datasets contain only RGB-NIR or RGB-TIR, while AG-ReID broadens viewpoints but lacks modality variety. Datasets that include both cross-modal and cross-view imagery [10, 29] typically provide limited per-identity scenario coverage and are evaluated by splitting into scenario-pair subsets, which hinders learning unified, scenario-invariant representations. Furthermore, daytime-centric collection undersamples nighttime conditions where infrared sensors are most valuable, limiting all-day robustness evaluation. Thus, ***II)*** *How to build a benchmark that captures diverse scenarios and real-world complexity to support a unified ReID task?*

On the methodology side, many approaches address predefined scenario pairs with multi-branch architectures [41, 57] or pairwise alignment losses [3, 44]. As modalities and viewpoints grow, such designs rely on scenario-specific modules or pairwise objectives, inflating parameters and scaling quadratically with scenarios, which is ill-suited for any-scenario settings. Moreover, heterogeneous sensing in AS-ReID introduces large appearance gaps across scenarios, while mixing all scenarios in a single gallery dramatically increases visually similar but semantically different distractors. As a result, AS-ReID poses two intertwined representation challenges: *(a) scenario-invariant intra-identity cohesion*, where features of the same identity should cluster across scenarios in a unified space; and *(b) large-margin inter-identity discrimination*, where features should maintain clear global margins amid abundant hard negatives from heterogeneous galleries. Accordingly, ***III)*** *How to design a single, scalable framework that learns effectively from heterogeneous sources while simultaneously enforcing (a) and (b) for any-scenario ReID?*

For problem ***I)***, we introduce a new task named **Any-Scenario Person ReID (AS-ReID)**. Here, a *scenario* refers to a combination of modality and viewpoint from heterogeneous sensors. The task requires that, given a query captured in any *scenario*, a single model retrieves the same identity from a heterogeneous gallery that spans images from all available *scenarios*. AS-ReID subsumes traditional single-scenario, cross-modal, and cross-view cases within a single task, thereby avoiding the closed-world assumption of training separate models for predefined scenario pairs and providing a unified, deployment-aligned training and evaluation paradigm.

For problem ***II)***, we contribute a large-scale multispectral aerial-ground benchmark **WHU-MARS**. WHU-MARS comprises 2,337 identities and 434,620 images captured by RGB-NIR-TIR sensors on both ground and UAV platforms, with tri-spectral, cross-view coverage for each identity, enabling unified training and evaluation for AS-ReID. WHU-MARS covers day and night, multiple seasons, and diverse weather, yielding broad variations in illumination, viewpoint, scale, and spectral characteristics that reflect real-world diversity. In addition, we provide a paired-only subset with synchronized tri-spectral samples, enabling consistent evaluation of AG-, VI-, MM-, and AS-ReID settings on the same identities. To the best of our knowledge, WHU-MARS is the largest multispectral person ReID dataset to date and provides a solid foundation for AS-ReID.

For problem ***III)***, we propose a **Unified Alignment and Discrimination (UAD)** framework to address two core challenges in AS-ReID: **(a) Progressive Center Alignment (ProCA)** consolidates view diversity into modality centers and aligns them toward identity centers to strengthen scenario consensus. **(b) Global Prototype Discrimination (GPD)** complements ProCA by injecting global structure, contrasting samples against global prototypes to impose large margins in the representation space. UAD trains a single, scenario-invariant model without extra branches or pairwise objectives and achieves strong performance in AS-ReID.

The main contributions are summarized as follows:

- **Unified ReID Task: AS-ReID.** We formulate Any-Scenario ReID (AS-ReID), a unified task that performs any-to-any retrieval across *scenarios* with a single model, aligned with real-world applications and subsuming conventional ReID tasks.
- **Multi-Protocol Benchmark: WHU-MARS.** We introduce a multispectral aerial-ground benchmark with 2,337 identities and RGB/NIR/TIR aerial/ground views, capturing diverse illumination, seasons, and weather conditions, and supporting AS-ReID and existing protocols.
- **Methodological Contribution.** We propose the Unified Alignment and Discrimination (UAD) framework, combining Progressive Center Alignment for scenario-invariant intra-identity cohesion with Global Prototype Discrimination for large-margin inter-identity separation.
- **Empirical Contribution.** We conduct extensive experiments under AG-, VI-, MM-, and AS-ReID protocols, which highlight the challenge of WHU-MARS and demonstrate the effectiveness of UAD for AS-ReID.

Table 1. Comparison of person ReID datasets across scenario settings. Counts under *Ground* and *Aerial* are per-modality image numbers. *Paired* indicates the availability of frame-synchronized multimodal paired samples. *Day/Night* indicate the presence of daytime/nighttime imagery. WHU-MARS provides tri-spectral, cross-view coverage for every identity, supporting AS-ReID and conventional ReID tasks.

Task	Dataset	Paired	Day	Night	#IDs	#Images	Ground			Aerial		
							#RGB	#NIR	#TIR	#RGB	#NIR	#TIR
VI-ReID	RegDB [27]	✓	✓	✓	412	8,240	4,120	-	4,120	-	-	-
	SYSU-MM01 [37]		✓	✓	491	45,863	30,071	15,792	-	-	-	-
	ThermalWorld [15]	✓	✓	✓	516	30,236	15,118	-	15,118	-	-	-
	LLCM [56]			✓	1,064	46,767	25,626	21,141	-	-	-	-
	AT-USTC [22]		✓	✓	270	135,465	69,791	65,674	-	-	-	-
AG-ReID	PRAI-1581 [53]		✓		1,581	39,461	-	-	-	39,461	-	-
	UAV-Human [20]		✓		1,144	41,290	-	-	-	41,290	-	-
	AG-ReID.V2 [28]		✓		1,615	100,502	53,723	-	-	46,779	-	-
	G2A-VReID [54]		✓		2,788	185,907	69,234	-	-	116,673	-	-
	LAGPeR [31]		✓	✓	4,231	63,841	∼44,000	-	-	∼21,000	-	-
VI & AG ReID	MP-ReID [10]		✓		1,930	136,156	45,270	36,300	-	26,046	-	28,543
	AG-VPReID.VIR [29]		✓		1,104	119,958	54,546	-	6,735	54,784	-	3,893
MM-ReID	RGBNT201 [59]	✓	✓	✓	201	14,361	4,787	4,787	4,787	-	-	-
AS-ReID and Above	**WHU-MARS-1000**	✓	✓	✓	1,000	**185,922**	46,212	46,212	**46,212**	15,762	**15,762**	15,762
	WHU-MARS-2337	✓	✓	✓	2,337	**434,620**	**119,636**	**117,512**	**88,847**	36,256	**38,143**	**34,226**

2. Related Work

2.1. Person Re-Identification Datasets

Across existing ReID tasks, numerous benchmarks exist, as summarized in Table 1. For VI-ReID, SYSU-MM01 [37] and RegDB [27] support visible-infrared all-day retrieval, while LLCM [56] and AT-USTC [22] emphasize challenging nighttime and long-term settings. To incorporate aerial viewpoints, PRAI-1581 [53] and UAV-Human [20] provide UAV-captured imagery, and AG-ReID.v2 [28] and G2A-VReID [54] extend to aerial-ground matching; CARGO [52] provides large-scale synthetic aerial-ground data. MP-ReID [10] and AG-VPReID.VIR [29] broaden coverage for cross-platform cross-modality evaluation, and RGBNT201 [59] offers tri-spectral paired samples for MM-ReID. A complementary line composes or extends existing datasets to widen scenario coverage: MMReID-Bench [17] aggregates multiple ReID tasks and datasets, and recent augmentations add sketches [2, 63] or textual descriptions [5, 13, 14] to classic ReID benchmarks. Collectively, these datasets provide solid task-specific foundations.

Despite spanning diverse scenarios, most datasets follow predefined scenario pairs and are evaluated via pair-specific splits, which constrain unified training and any-to-any evaluation. Cross-source compositions introduce domain and protocol shifts and rarely provide consistent multi-scenario observations per identity. They also remain largely daytime-centric, undersampling nighttime conditions. These factors motivate a benchmark for real-world applications: consistent identity observations across modalities and viewpoints under day-night, seasonal, and weather variations, enabling unified any-to-any retrieval and compatible with standard VI-, AG-, and MM-ReID protocols.

2.2. Multi-Scenario ReID Methods

Recent efforts aim to model diverse scenarios within a single network. Prompt-based approaches such as Uni-Prompt [10], Instruct-ReID [12], and VersReID [60] learn scenario- or task-specific prompts to guide multi-scenario retrieval. Unified-representation methods [16, 42, 63] encode heterogeneous modalities in a shared space. Task-focused extensions broaden coverage: Uni-AT [22] targets anytime retrieval with scenario-specific class tokens; MixER [1] and MixReID [23] extend VI-ReID to both cross- and intra-modality retrieval; MCL [45] explores unpaired USL-VI-ReID; and MDReID [7] and UCPL [62] support modality mismatch and cross-modality retrieval in MM-ReID. Together, these lines of work extend multi-scenario modeling via prompt conditioning, unified representation, and task-specific protocol extensions.

Despite this progress, current approaches remain insufficient for AS-ReID. VI-style solutions often adopt multi-branch architectures or pairwise objectives, resulting in complexity that grows quickly as scenarios increase. Prompt-based systems commonly train separate models for predefined scenario pairs or on heterogeneous datasets with disjoint identities, which limits learning identity-consistent features across scenarios. Unified-representation pipelines are optimized with RGB-anchored objectives, favoring visible galleries rather than symmetric any-to-any matching across modalities and viewpoints. Protocol extensions that attach separate heads or scenario-specific retrieval rules further partition the feature space and complicate scaling to heterogeneous galleries. These limitations motivate a simple but effective framework that learns a unified, scenario-irrelevant representation with a single model for AS-ReID.

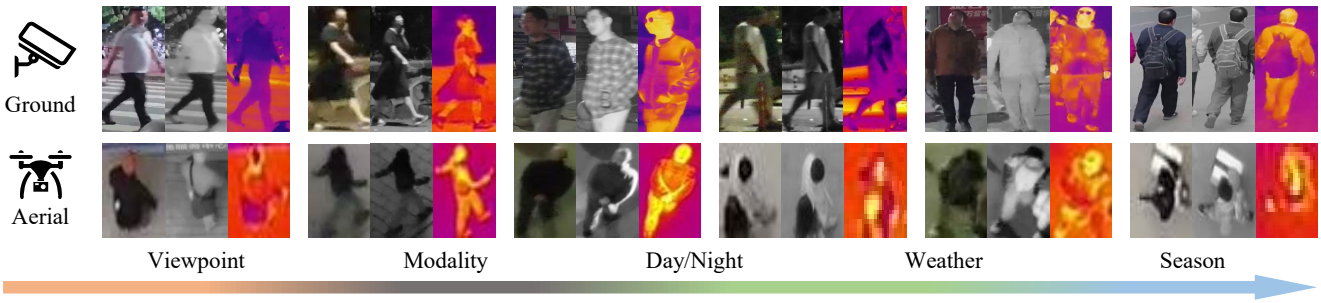


Figure 2. Examples from the WHU-MARS dataset. Top and bottom rows show images captured by ground and aerial platforms, respectively. Each group shows paired tri-spectral samples, illustrating variations across viewpoint, modality, day/night, weather, and season.

3. The WHU-MARS Dataset

3.1. Dataset Overview

We introduce WHU-MARS, a large-scale multispectral aerial-ground ReID dataset designed for AS-ReID, which also supports standard AG-, VI-, and MM-ReID protocols. Data were collected on a university campus using two UAV platforms and five ground camera nodes. The UAV platforms carried DJI H20T cameras at altitudes between 20-50m, while the ground nodes employed custom tri-spectral cameras at about 1.5m. Both platforms synchronously recorded RGB, NIR, and TIR video streams. The collection spanned from July to the following January between 09:00 and 21:00 local time, covering both daytime and nighttime and capturing diverse seasonal and weather conditions. Figure 2 shows representative samples highlighting the diversity of WHU-MARS. WHU-MARS contains 2,337 identities and 434,620 images across 13 sessions, totaling over 38 hours of video. Each identity appears in all three modalities and multiple views. As summarized in Table 1, to our knowledge, WHU-MARS is the largest multispectral person ReID dataset to date. Data collection was authorized by the institutional office and conducted in publicly accessible areas. On-site notices were posted during capture, and all images underwent automatic face mosaicking and contain no personally identifiable metadata to protect privacy. Additional device specifications, annotation procedures, and detailed statistics are provided in the supplementary material.

3.2. Dataset Features

Scale and Tri-Spectral Coverage. WHU-MARS comprises 2,337 identities and 434,620 images, and every identity is observed in RGB, NIR, and TIR across views, enabling AS-ReID and supporting standard AG-, VI-, and MM-ReID protocols within a single benchmark.

Day-Night Imaging. The collection spans both daytime and nighttime. Diurnal variations cause pronounced changes in illumination and background, increasing retrieval difficulty and encouraging models to learn time-invariant identity information.

Table 2. Statistics of the two WHU-MARS versions.

Version	Split	#IDs	#Images	#RGB	#NIR	#TIR
WHU-MARS-2337	Train	1,000	217,504	77,253	76,975	63,276
	Gallery	1,337	217,116	78,639	78,680	59,797
	Query	1,337	15,020	5,254	5,275	4,491
WHU-MARS-1000	Train	500	92,313	30,771	30,771	30,771
	Gallery	500	93,609	31,203	31,203	31,203
	Query	500	6,405	2,135	2,135	2,135

Seasonal and Weather Diversity. The dataset covers multiple seasons and varied weather, inducing changes in ambient temperature, backgrounds, and clothing. These factors affect both visible appearance and thermal emission patterns, yielding realistic intra-identity variation.

Multimodal Heterogeneity. Multiple sensor models and acquisition settings were used for RGB, NIR, and TIR, inducing cross-spectral distribution shifts and broader intra-modality variation representative of real-world applications.

Viewpoint and Scale Variation. Ground cameras and UAV platforms at different heights and oblique angles provide diverse viewpoints, subject scales, and image resolutions, mirroring the challenges of multi-platform surveillance.

3.3. Evaluation Protocol

In tri-spectral systems, differences in sensor resolution and aspect ratio, as well as occasional occlusions, mean that some images do not have frame-synchronized RGB-NIR-TIR counterparts. This characteristic reflects real-world deployments and aligns well with the AS-ReID setting. However, tasks that require paired samples such as MM-ReID cannot utilize the full dataset under such constraints. Moreover, scaling to 2,337 identities and over 400K images imposes substantially higher computational cost than common ReID benchmarks. To address these considerations, we define two official versions of WHU-MARS: WHU-MARS-2337 and WHU-MARS-1000.

WHU-MARS-2337 retains all available images and identities and serves as the full benchmark for challenging AS-ReID evaluation and multispectral pretraining. WHU-MARS-1000 is a constructed subset of 1,000 identities with

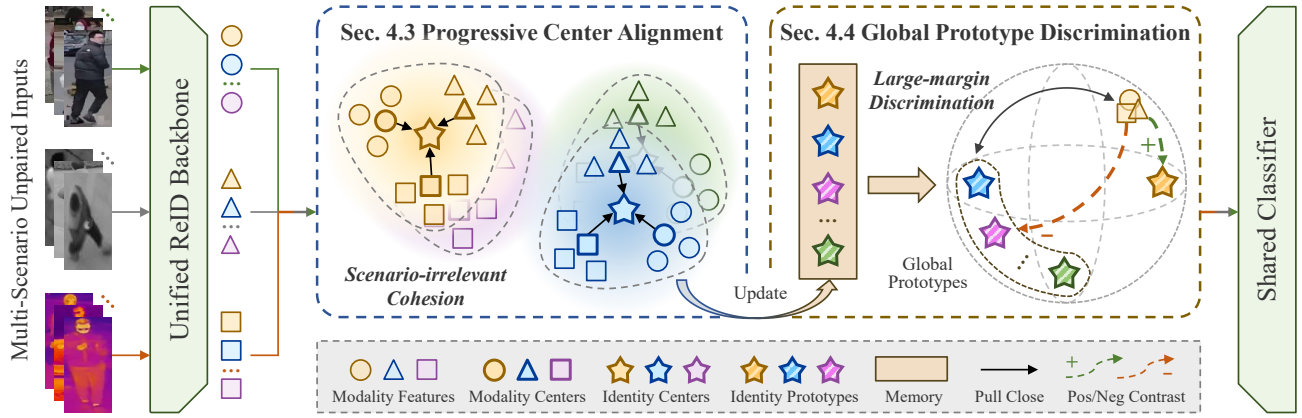


Figure 3. Overview of the Unified Alignment and Discrimination (UAD) framework for AS-ReID. Colors encode identities; shapes encode modalities. Multi-scenario unpaired inputs (RGB/NIR/TIR, aerial/ground) are processed by a single model. Progressive Center Alignment (ProCA) consolidates view diversity into modality centers and then pulls these centers toward an identity center, strengthening scenario-irrelevant intra-identity cohesion. The identity centers (stars) update a momentum prototype memory to form global prototypes. Global Prototype Discrimination (GPD) contrasts features against these global prototypes to enforce large inter-identity discrimination.

61,974 frame-synchronized tri-spectral triplets. It enables unified benchmarking at a moderate scale, providing identical data for AG-, VI-, MM-, and AS-ReID tasks. For identity alignment across scales, the train/test identities of WHU-MARS-1000 are subsets of the corresponding splits in WHU-MARS-2337. Query sets are formed by randomly selecting 1-3 images per identity per camera from the respective galleries. Detailed statistics are reported in Table 2.

Aligned with established practice, we report the Cumulative Matching Characteristic (CMC) curve and mean Average Precision (mAP). For AS-ReID, we exclude all gallery images from the same camera as the query, following common VI-ReID protocols. For other tasks, we adopt their standard, task-specific evaluation protocols.

4. Methodology

We first formalize the proposed Any-Scenario ReID (AS-ReID) task, then present a unified single-branch baseline tailored to this setting, and finally introduce the Unified Alignment and Discrimination (UAD) framework. UAD targets the two challenges highlighted in AS-ReID: (i) *scenario-irrelevant intra-identity cohesion* and (ii) *strong inter-identity discrimination* under large heterogeneous galleries. Figure 3 illustrates the overall architecture.

4.1. Task Definition: Any-Scenario ReID

Scenarios. Let $\mathcal{M} = \{\text{RGB}, \text{NIR}, \text{TIR}, \ldots\}$ be the set of modalities and $\mathcal{V} = \{\text{ground}, \text{aerial}, \ldots\}$ be the set of viewpoints. A *scenario* is a modality-viewpoint combination, and the *scenario* space is the Cartesian product

$$\mathcal{S} = \mathcal{M} \times \mathcal{V}. \tag{1}$$

Data and single model. Let $\mathcal{X}$ denote the image space and $\mathcal{Y}$ the identity set. Each image x with (y, s, c) where $y \in \mathcal{Y}$ is the identity, $s \in \mathcal{S}$ the scenario, and c the camera id. AS-ReID uses a *single model* $f_\theta : \mathcal{X} \to \mathbb{R}^d$ shared across all scenarios to map images into one unified space.

Evaluation protocol. Let $\mathcal{Q}$ be the query set and G be the gallery set spanning all scenarios. For a query $q = (x_q, y_q, s_q, c_q) \in \mathcal{Q}$, the evaluation gallery is

$$g_q = \{(x, y, s, c) \in G \mid c \neq c_q\}. \tag{2}$$

We rank g_q by embedding-space distance with $f_\theta(x_q)$. The objective is to learn θ such that items in g_q with identity y_q are ranked above those with $y \neq y_q$. Classical subtasks (VI-, AG-, MM-, Tr-ReID) are recovered by restricting $\mathcal{S}$ and filtering $\mathcal{Q}$ and G accordingly.

4.2. Unified Baseline for AS-ReID

We adopt a simple unified single-branch baseline for AS-ReID. Unlike MM-ReID settings that rely on synchronized paired tri-spectral samples, AS-ReID makes no pairing assumption. To expose the model to heterogeneous conditions within each mini-batch, we adopt a $P \times M \times K$ sampler, where P is the number of identities, $M = |\mathcal{M}|$ is the number of modalities, and K is the number of images per identity per modality. This yields batches with rich cross-modality and cross-view variability per identity.

All images are processed by a single backbone f_θ shared across scenarios. Given an input x, we obtain a feature $f = f_\theta(x)$ and a BNNeck [25] feature $z = \text{BN}(f)$. The BN layer and classifier are shared to encourage learning in a unified space. At inference, we ℓ_2-normalize f for retrieval.

Following common practice, we apply the Weighted Regularization Triplet loss $\mathcal{L}_{WRT}$ [47] to f for metric learning and the cross-entropy loss $\mathcal{L}_{ce}$ to $\mathbf{z}$ for classification:

$$\mathcal{L}_{base} = \mathcal{L}_{ce} + \mathcal{L}_{WRT}. \tag{3}$$

This single-branch baseline serves as a simple, scenario-irrelevant reference for AS-ReID.

4.3. Progressive Center Alignment

Heterogeneous sensing in AS-ReID induces scenario-specific biases: features of the same identity may disperse across modalities and viewpoints, harming cross-scenario retrieval. Rather than aligning scenario pairs, which scales quadratically in the number of scenarios or requires image-level pairing, we adopt a two-level **Pro**gressive **C**enter **A**lignment (ProCA) to mitigate scenario shifts and promote intra-identity cohesion in the unified representation space.

Given a $P \times M \times K$ mini-batch as described in Sec. 4.2, let $\{f_{m,k}^{y}\}_{k=1}^{K}$ be the ℓ_2-normalized embeddings. ProCA first aggregates multiple views into a modality center, which implicitly attenuates view-specific noise and scale changes:

$$\tilde{\boldsymbol{\mu}}_{y,m} = \frac{1}{K}\sum_{k=1}^{K} f_{m,k}^{y}, \qquad \boldsymbol{\mu}_{y,m} = \mathrm{norm}(\tilde{\boldsymbol{\mu}}_{y,m}). \tag{4}$$

ProCA then aggregates modality centers into an identity anchor, where $\mathrm{norm}(\cdot)$ denotes ℓ_2 normalization, and the identity center $\boldsymbol{\mu}_y$ is treated as a stop-gradient anchor:

$$\tilde{\boldsymbol{\mu}}_{y} = \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}} \boldsymbol{\mu}_{y,m}, \qquad \boldsymbol{\mu}_{y} = \mathrm{norm}(\tilde{\boldsymbol{\mu}}_{y}). \tag{5}$$

Finally, ProCA minimizes the identity-wise between-scenario scatter by pulling each modality center toward its identity anchor:

$$\mathcal{L}_{ProCA} = \frac{1}{P}\sum_{y\in\mathcal{B}} \frac{1}{|\mathcal{M}|}\sum_{m\in\mathcal{M}} \left\|\boldsymbol{\mu}_{y,m} - \boldsymbol{\mu}_{y}\right\|_2^2. \tag{6}$$

This progressive design first consolidates multi-view variability within each modality and then aligns modalities at the identity level, yielding lower identity-wise between-scenario scatter and providing stable anchors for GPD. The formulation is pairing-free, scales linearly with scenarios, and integrates naturally with the unified training paradigm.

4.4. Global Prototype Discrimination

Retrieval over a large heterogeneous gallery in AS-ReID produces abundant distractors (e.g., same-modality or same-view hard negatives), demanding strong inter-identity discrimination. While metric losses such as triplet enlarge margins within a mini-batch, they shape geometry only locally. We therefore introduce **G**lobal **P**rototype **D**iscrimination (GPD), which complements ProCA by injecting global structure at every iteration and maximizing dataset-level discrimination under heterogeneous negatives.

Let $\mathcal{P}^{(t)} = \{\mathbf{p}_y \in \mathbb{R}^d \mid y \in \mathcal{Y}_{\text{train}}\}$ denote an identity-level, ℓ_2-normalized prototype memory at iteration t. Within iteration t, the prototype memory is first kept fixed to perform global prototype contrast. Afterward, the memory is updated using batch-wise identity centers $\boldsymbol{\mu}_y$ from Eq. (5) via exponential moving average:

$$\mathbf{p}_y \leftarrow \mathrm{norm}\big(\alpha\,\mathbf{p}_y + (1-\alpha)\,\boldsymbol{\mu}_y\big), \tag{7}$$

where $\alpha \in [0, 1)$ is the momentum. This accumulates information across iterations without full-dataset recomputation. ProCA produces less noisy identity centers $\boldsymbol{\mu}_y$ for updating the prototype memory, while the global supervision from GPD encourages identity centers to be further apart in the embedding space to complement the aggregation of ProCA.

For the contrast objective, each normalized feature f_i takes its identity prototype $\mathbf{p}_{y_i}$ as the positive and all other prototypes $\{\mathbf{p}_y : y \neq y_i\}$ as negatives:

$$\mathcal{L}_{GPD}^{(i)} = -\log \frac{\exp(f_i^{\top}\mathbf{p}_{y_i}/\tau)}{\sum_{y\in\mathcal{Y}_{\text{train}}} \exp(f_i^{\top}\mathbf{p}_y/\tau)}, \tag{8}$$

where τ is the temperature and all vectors are ℓ_2-normalized. The loss over a mini-batch $\mathcal{B}$ is

$$\mathcal{L}_{GPD} = \frac{1}{|\mathcal{B}|}\sum_{i\in\mathcal{B}} \mathcal{L}_{GPD}^{(i)}. \tag{9}$$

GPD enforces near-uniform separation from all other identities by contrasting each sample against all identity prototypes. It complements local metric losses with dataset-level margins, strengthening large-margin discrimination over the global structure.

4.5. Overall Objective

Consistent with the retrieval ranking protocols, ProCA and GPD are applied to the pre-BN embedding $\mathbf{f}$. The final training objective of UAD is

$$\mathcal{L}_{UAD} = \mathcal{L}_{base} + \lambda_{\text{ProCA}}\,\mathcal{L}_{ProCA} + \lambda_{\text{GPD}}\,\mathcal{L}_{GPD}, \tag{10}$$

where λ_{ProCA} and λ_{GPD} are the weights for $\mathcal{L}_{ProCA}$, $\mathcal{L}_{GPD}$.

5. Experiment

5.1. Experimental Settings

Datasets. All experiments are conducted on the WHU-MARS introduced in Section 3. Unless otherwise specified, we use the WHU-MARS-1000 as the default and additionally use the WHU-MARS-2337 to assess scalability.

Evaluation Protocols. We report CMC (Rank-*k*) scores and mAP for all tasks. Following the standard VI-ReID protocol, we additionally report mean Inverse Negative Precision (mINP) [47] for VI-ReID only.

Implementation details. We adopt Vision Transformer [4] (ViT-Base) backbone pre-trained on ImageNet and train with Distributed Data Parallel (DDP) on four NVIDIA RTX 3090 GPUs. Images are resized to 128×256 and augmented with padding, random resized cropping, horizontal flipping, and random erasing [61]. We employ a $P\times M\times K$ sampler with P=16 identities, M=3 modalities, and K=4 images, yielding a per-GPU batch size of 192. The base learning rate is set to 0.008 per GPU, and optimization uses SGD for 120 epochs with a 5-epoch linear warm-up followed by cosine decay. For GPD, the momentum α is 0.8 and the temperature τ is 0.03. λ_{ProCA} and λ_{GPD} are set to 0.01, 1.0.

Task-specific settings. For a fair comparison with prior work, inputs are resized to 144×288 for VI-ReID and to

Table 3. Comparison under the AS-ReID setting. *WHU-MARS-1000* and *WHU-MARS-2337* are trained and evaluated on their respective splits. *WHU-MARS-1000-GD* evaluates the WHU-MARS-1000-trained model using ground and daytime queries and galleries only.

Method	Venue	Params	WHU-MARS-1000-GD				WHU-MARS-1000				WHU-MARS-2337			
			mAP	R-1	R-5	R-10	mAP	R-1	R-5	R-10	mAP	R-1	R-5	R-10
BoT [25]	CVPRW19	23.5M	10.6	24.9	37.9	45.6	5.9	18.4	32.5	40.1	4.6	14.8	26.2	32.5
AGW [47]	TPAMI21	23.5M	6.1	17.0	27.8	35.9	4.0	15.6	28.7	36.0	5.2	16.9	28.7	35.1
MSINet [9]	CVPR23	2.3M	9.7	26.9	39.5	46.3	5.6	21.1	37.5	46.0	2.6	14.8	25.4	31.6
TransReID [11]	ICCV21	99.9M	9.6	18.9	34.1	43.4	7.4	17.1	30.3	38.2	5.5	13.1	23.4	29.1
TransReID-SSL [26]	arXiv21	88.4M	14.0	30.1	44.1	51.7	10.1	26.7	43.8	51.7	9.2	24.4	39.1	46.2
DC-Former [21]	AAAI23	85.8M	9.9	20.1	34.9	42.0	6.5	16.3	29.9	36.7	4.3	10.9	19.8	25.5
CLIP-ReID [18]	AAAI23	125.3M	13.8	28.9	42.7	49.6	10.6	26.6	42.6	50.4	9.3	24.1	37.5	44.0
PHA [50]	CVPR23	107.0M	9.6	22.0	33.9	42.3	7.5	15.9	28.2	35.3	5.5	12.0	21.1	26.3
VDT [52]	CVPR24	85.8M	13.1	31.0	44.1	51.1	7.8	25.2	41.9	50.4	6.3	20.0	33.5	40.4
DTST [32]	arXiv24	86.9M	11.6	29.8	42.5	47.9	6.8	23.4	39.8	47.9	7.5	23.5	37.9	45.3
SeCap [31]	CVPR25	130.9M	13.4	28.1	41.2	47.8	10.4	26.4	42.4	50.4	8.0	21.4	34.1	41.0
UAD (Ours)	**CVPR26**	85.7M	**14.5**	**31.9**	**45.0**	**52.5**	**11.0**	**29.5**	**46.3**	**54.8**	**9.6**	**25.7**	**40.9**	**48.2**

128×256 for other tasks, and patch strides are set to 16×16 for transformer-based methods. All remaining settings follow the corresponding open-source implementations.

5.2. Experimental Analysis

In this section, we present a comprehensive evaluation on WHU-MARS across four settings, including AS-ReID, AG-ReID, VI-ReID, and MM-ReID, and analyze the impact of our key design choices on performance.

Any-Scenario ReID. We evaluate a range of recent general ReID methods under the AS-ReID protocol on WHU-MARS. Heterogeneous galleries that mix modalities and viewpoints make AS-ReID substantially more challenging than conventional settings. As detailed in Table 3, standard CNN and Transformer based methods such as BoT [25] and TransReID [11] achieve Rank-1 below 20% on WHU-MARS-1000. Human-centric pre-training and stronger backbones improve robustness: TransReID-SSL [26] and CLIP-ReID [18] consistently outperform other generic methods. AG-ReID methods that explicitly learn viewpoint invariance can partially transfer their benefits to AS-ReID and bring further gains, although cross-modality discrepancies still cap performance. Restricting evaluation to the ground-daytime subset WHU-MARS-1000-GD improves all methods, indicating that excluding nighttime and aerial viewpoints eases retrieval. However, the task remains nontrivial because modality gaps remain significant. Scaling from WHU-MARS-1000 to the larger and more diverse WHU-MARS-2337 leads to a consistent drop across methods, reflecting the increased number of hard negatives and broader domain shifts with more identities and scenarios. Across these settings, our UAD achieves the best overall performance and uses the fewest params among Transformer-based methods, indicating that learning a unified representation in a single model is feasible and competitive for AS-ReID while remaining practical for deployment. We attribute the gains primarily to ProCA and GPD, which jointly enforce scenario-invariant cohesion and global discrimination under heterogeneous galleries.

Table 4. Comparison under the AG-ReID setting.

Setting	A→G			G→A		
	mAP	R-1	R-10	mAP	R-1	R-10
VDT [52]	8.9	30.4	56.7	10.7	30.2	56.4
DTST [32]	7.9	30.5	56.2	9.8	28.5	**57.3**
SeCap [31]	10.1	26.4	50.4	12.0	25.3	50.7
UAD (Ours)	**11.5**	**31.9**	**58.3**	**13.3**	**30.5**	54.7

Table 5. Comparison under the VI-ReID setting.

Method	NIR→RGB			TIR→RGB		
	R-1	mAP	mINP	R-1	mAP	mINP
AGW [47]	17.42	10.87	1.49	5.67	3.50	0.60
CAJ [46]	25.90	16.11	2.04	7.92	4.79	0.75
CAJ+ [48]	25.57	14.62	1.68	7.78	4.37	0.66
SAAI [6]	23.47	13.92	1.58	5.81	3.57	0.65
CSDN [49]	25.11	14.61	2.07	6.89	3.84	0.73
DEEN [56]	**27.63**	17.05	3.00	8.01	5.69	1.31
PMT [24]	27.12	**18.16**	**3.18**	8.29	5.85	1.21
UAD (Ours)	25.53	16.58	3.10	**9.04**	**6.32**	**1.36**

Aerial-Ground ReID. We compare AG-ReID specific methods and also report the cross-view performance of UAD. All methods are trained on WHU-MARS-1000 and evaluated with queries and galleries restricted to the corresponding viewpoints. As shown in Table 4, SeCap [31] attains the best mAP among AG-specific approaches, while VDT [52] and DTST [32] achieve comparable Rank-1 accuracy. Despite not being tailored to AG-ReID, UAD achieves the strongest cross-view results, suggesting that a unified representation generalizes well across viewpoints and mitigates severe scale and perspective changes.

Visible-Infrared ReID. We evaluate WHU-MARS-1000 under VI-ReID protocols. As summarized in Table 5, all specialized VI-ReID methods are trained separately with two independent models, whereas UAD is trained once on all three modalities without any modality-specific branches, pairwise alignment losses, or VI-specific augmentation (CA [46] or RLE [30]). All images are resized to 144×288. Compared with previous VI-ReID benchmarks, all methods obtain much lower performance

Table 6. Modality-pair evaluation of the single UAD model. Columns and rows indicate the query and gallery modality.

G / Q	RGB			NIR			TIR		
	R-1	mAP	mINP	R-1	mAP	mINP	R-1	mAP	mINP
RGB	40.47	26.83	5.36	26.56	16.52	2.43	8.10	6.00	1.11
NIR	25.53	16.58	3.10	29.79	17.93	2.64	7.78	5.87	1.11
TIR	9.04	6.32	1.36	8.57	5.69	1.02	20.98	14.00	2.35

Table 7. Comparison under the MM-ReID setting.

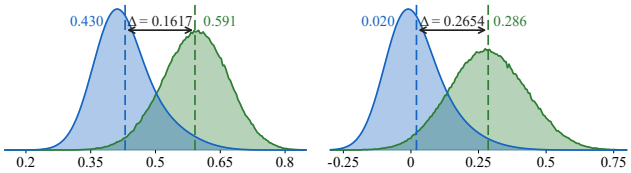
Method	mAP	R-1	R-5	R-10
ICPL-ReID [19]	35.0	43.5	59.2	67.5
TOP-ReID [33]	36.2	44.0	61.5	69.5
EDITOR [51]	37.8	44.9	63.0	72.0
HTT [36]	39.1	48.0	64.1	72.6
MDReID [7]	41.4	52.1	67.7	74.8
MambaPro [34]	42.1	53.7	69.7	77.0
MFRNet [8]	43.8	54.3	70.0	77.1
PromptMA [55]	44.2	53.7	68.4	75.1
DeMo [35]	**44.8**	**56.2**	**72.6**	**78.7**

Table 8. Ablation study of the UAD framework.

$\mathcal{L}_{base}$	ProCA	GPD	mAP	R-1	R-5	R-10
✓			8.9	23.9	41.4	50.5
✓	✓		9.3	24.5	42.0	50.7
✓		✓	10.9	28.7	45.9	54.2
✓	✓	✓	**11.0**	**29.5**	**46.3**	**54.8**

on WHU-MARS-1000, indicating that the multi-view, heterogeneous camera setting is considerably more challenging. On NIR-RGB, PMT [24] achieves the best overall performance, while our unified UAD model remains competitive among recent methods despite not being tailored to VI-ReID. In contrast, on the more difficult TIR-RGB protocol, UAD attains the best results among all competitors. This suggests that any-scenario training with ProCA and GPD enforcing shared identity prototypes helps the scarce, noisy TIR modality benefit from RGB/NIR supervision. Table 6 further evaluates the same UAD model on all 3×3 query-gallery modality pairs: intra-modality retrieval is easiest and TIR to others are hardest, yet the unified model still yields reasonable performance, demonstrating that a single unified model without modality-specific design can handle multiple VI-ReID tasks competitively.

Multi-modality ReID. We conduct a comprehensive benchmark on WHU-MARS-1000 for publicly available MM-ReID methods in Table 7. In this setting, all methods are trained and evaluated with synchronized tri-spectral triplets, allowing them to exploit complementary cues across modalities, which is strictly easier than the AS-ReID setting. Nevertheless, ViT-based methods achieve less than 50% Rank-1, and the best method DeMo [35] reaches only 44.8% mAP and 56.2% Rank-1, indicating that WHU-MARS remains a substantially more challenging multi-modality ReID benchmark than existing datasets.



(a) Baseline (b) UAD

Figure 4. Visualization of cosine similarity distributions for intra- and inter-identity pairs. Green and blue denote intra- and inter-identity pairs, and dashed lines indicate the mean cosine similarity.

5.3. Ablation Study

We conduct ablations under the AS-ReID protocol to assess the contribution of each component in UAD. As shown in Table 8, introducing ProCA leads to consistent improvements, indicating that progressive alignment strengthens scenario-irrelevant learning. Enabling GPD produces larger gains by injecting the global prototype structure. Combining ProCA and GPD achieves the best overall performance, confirming the intended synergy: ProCA progressively regularizes the batch identity centers and stabilizes the prototype memory, while GPD leverages them to sharpen global discrimination under heterogeneous galleries.

5.4. Visualization

To further examine how UAD shapes the representation space, we visualize the cosine similarity distributions of intra-identity and inter-identity pairs in Figure 4. Compared with the baseline, which exhibits substantially concentrated and overlapped distributions under scenario heterogeneity, UAD widens the mean similarity margin between the two distributions, indicating a more separable and discriminative representation space for robust retrieval.

6. Conclusion

We introduce Any-Scenario ReID (AS-ReID), a unified and deployment-aligned paradigm in which a single model retrieves identities from a heterogeneous gallery across diverse scenarios. To support this task, we contribute WHU-MARS, a multispectral aerial-ground dataset with tri-spectral, cross-view coverage for each identity, supporting AS-ReID as well as conventional AG-, VI-, and MM-ReID protocols. Methodologically, our Unified Alignment and Discrimination (UAD) framework strengthens scenario-invariant intra-identity cohesion through Progressive Center Alignment (ProCA) and enforces large inter-identity margins via Global Prototype Discrimination (GPD). Extensive experiments under multiple protocols highlight the challenge of WHU-MARS and validate the effectiveness of UAD. We believe WHU-MARS provides a solid data foundation for the ReID community, and that AS-ReID and UAD represent concrete steps toward more general and deployment-friendly person ReID systems.

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